

Water Electrolysis

Lecture & Instructor Notes

Audience: instructors and teaching assistants

Purpose: Teaching script, deep-dive notes, and coaching guidance

Session hook: Predict a gas volume, then go and measure it

Length: 90-minute lab block (~20 min concepts + ~70 min hands-on)

This packet includes

1. Session overview & plan
2. Lecture talking points
3. Instructor deep-dive notes
4. Preparation reminders

Generated from the Electrochemistry workshop curriculum. Prefer the latest PDF from the course repository if procedures change mid-week.

Session 4 — Water Electrolysis and Gas Collection

Hook: Pure water barely conducts. Add one spoonful of salt-that-isn't-table-salt and the same water tears itself into hydrogen and oxygen — in a ratio you can measure with a ruler.

Learning outcomes

- Explain why pure and tap water electrolyze poorly (**too few mobile ions**)
- Show that Na_2SO_4 raises current and therefore reaction rate, without being consumed
- Measure **current**, and use Faraday's law to **predict a gas volume before collecting it**
- Relate applied **voltage** to the ~ 1.23 V thermodynamic floor and to overpotential
- Collect H_2 and O_2 separately in the **U-tube** and confirm the **2:1** ratio
- Identify both gases by their **splint tests**, and hydrogen again by a balloon pop
- **Start** the electrolytic rust-removal cells that Session 5 opens with

Session plan (90 min)

Time	Activity	Reference
-5 min	TA: switch on the balloon H_2 generator. It must run the whole session — see the timing table in experiment.md Part D	preparation.md
0-8 min	Review Days 1-3; hook: two rods in pure water, nothing happens	lecture.md §1-3
8-18 min	Concepts: ions carry current, supporting electrolyte, 1.23 V, $\dot{n} = I/(nF)$, 2:1	lecture.md §3
18-26 min	Quiz check + pre-lab prediction (required before Part C)	lecture.md §4-5
26-36 min	Part A — Conductivity ladder: distilled $\rightarrow$ tap $\rightarrow$ electrolyte	experiment.md — Part A
36-50 min	Part B — Measure I ; calculate $\dot{n}$; write a volume prediction	experiment.md — Part B
50-72 min	Part C — U-tube collection; ratio and Faraday check	experiment.md — Part C
72-82 min	Part D — Splint tests, then the balloon pop	experiment.md — Part D
82-90 min	Part E — Start derusting cells for Session 5	experiment.md — Part E

The one thing that ruins this session: starting the balloon generator when you reach Part D. At a realistic classroom current it needs **30-60 minutes** to fill. Start it before the students walk in.

Key reactions

Overall: $2\text{H}_2\text{O}(\text{l}) \rightarrow 2\text{H}_2(\text{g}) + \text{O}_2(\text{g})$

Cathode (-): $2\text{H}_2\text{O} + 2\text{e}^- \rightarrow \text{H}_2 + 2\text{OH}^-$

Anode (+): $2\text{H}_2\text{O} \rightarrow \text{O}_2 + 4\text{H}^+ + 4\text{e}^-$ (acidic form)
or $4\text{OH}^- \rightarrow \text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^-$ (basic form)

Rate: $\dot{n}(\text{H}_2) = I/(2F)$ $\dot{n}(\text{O}_2) = I/(4F)$

Rule of thumb: 1 ampere makes ≈ 7.5 mL of H_2 per minute at room conditions.

Status

- ☐ Balloon H_2 generator tested and its fill time measured
- ☐ ~ 0.1 M Na_2SO_4 mixed and labeled "**NO SALT**"
- ☐ U-tube pilot-tested: fill level correct, ratio visible inside 20 min
- ☐ U-tube internal diameter measured and written on the board
- ☐ Splints and a lighter staged; balloon area cleared
- ☐ **No NaCl** anywhere in this room today except the sealed Session 5 corrosion jars
- ☐ Rusty nails, washing soda, and a **mains-powered supply** ready for Part E

Session 4 — Lecture Notes: Water Electrolysis and Stoichiometry

Audience: instructors / TAs — use this as your teaching script. **Students** get the talking points during class and complete the \$5 prediction sheet before the U-tube.

Deep-dive chemistry, overpotential, and coaching traps: instructor.md. Bench procedure: experiment.md.

1. Summary of concepts from prior days

Spend **5-8 minutes** here. Today leans on all three previous sessions.

Session 1 — Fruit battery (galvanic cell)

Idea	What to remind them
Spontaneous chemistry	The reaction made electricity by itself
Electron vs ion path	Electrons in the wire, ions in the juice — both are needed
Weak electrolyte	Fruit juice conducted poorly, so the current was feeble

Session 2 — Copper electroplating (electrolytic cell)

Idea	What to remind them
Electrolysis	Electricity forces a reaction that would not happen on its own
Polarity flips	Object to plate goes to battery (-) = cathode = reduction
Active anode	The copper anode dissolved and resupplied Cu^{2+}

Session 3 — Silver plating and Faraday's law

Idea	What to remind them
$Q = I \times t$	Charge is current multiplied by time
$n_e = Q/F$	$F = 96,485 \text{ C}$ per mole of electrons
Electrons per atom	Ag needed 1 e^- ; Cu needed 2. The number matters
Inert anode	Graphite did not dissolve and did not resupply Ag^+
Current efficiency	We <i>assumed</i> 100% and could not really check it

One-sentence bridge: *"For three days the thing being changed was a metal. Today it is the water itself."*

What is new today

- The substance being split is the **solvent**, not something dissolved in it
- The products are **gases**, which means for the first time we can **measure** how much we made

- Two products, with **different electron counts** — 2 e^- per H_2 , 4 e^- per O_2 — so the amounts must come out in a fixed ratio
- The salt we add is **not consumed**. It is a supporting electrolyte, a new idea

One-sentence goal: *"Today we predict how much gas we will make before we make it — and then we check."*

2. Class plan (90 min)

Time	Block	What you do	Student product
-5 min	TA starts the balloon generator	See experiment.md Part D	—
0-8 min	Review + hook	§1 and §3.1; two rods in distilled water, nothing happens	Oral answers
8-18 min	Teach today's concepts	§3.2-3.6; board flowchart	Notes / vocab
18-26 min	Quiz check + pre-lab	§4 questions, then §5 prediction sheet	Completed prediction sheet
26-36 min	Part A — conductivity ladder	Fixed voltage, three liquids	Ladder table filled
36-50 min	Part B — measure I, predict volume	Faraday arithmetic with their current	Predicted mL of H_2
50-72 min	Part C — U-tube	Ratio and prediction check	Gas log + efficiency %
72-82 min	Part D — splint tests, balloon	Instructor demo	Gas identifications
82-90 min	Part E — start derusting	Polarity called out as a class	Cells running

Pacing tip: the prediction sheet in §5 is the gate. A group that has not written down a predicted volume has nothing to compare their U-tube data with, and Part C collapses into watching bubbles.

3. Detailed explanations to give students

3.1 Hook (say this)

Hold up a beaker of distilled water with two graphite rods in it, already wired to the supply.

"This is pure water. Water is H_2O — the hydrogen and oxygen are already in there, we are not making anything exotic. I have several volts across these two rods. Watch."

Switch on. Nothing happens.

"Nothing. So what is missing? The reactant is right there. The voltage is right there. What else does a chemical reaction driven by electricity need?"

Take answers. Steer toward: **charge has to be able to get across the gap**, and in a liquid that means **ions have to move**.

3.2 Why pure water barely conducts (board)

Water does ionize, a little:



so $[\text{H}^+] = 10^{-7}$ M. Put that next to the concentration of water itself, 55.5 M:

About one molecule in 550 million is ionized at any moment.

That is the whole of Part A rung 1. There is no road for the charge.

Tap water is better — dissolved minerals give it real ions — but still hundreds of times worse than what we are about to make.

3.3 The supporting electrolyte — a genuinely new idea

Dissolve Na_2SO_4 and the solution fills with Na^+ and SO_4^{2-} . Current can now cross.

But here is the part worth slowing down for, because it is not obvious and students will get it wrong on a test:

Neither Na^+ nor SO_4^{2-} reacts. Sodium is far too hard to reduce and sulfate is already fully oxidized. They ferry charge through the solution and are exactly as abundant at the end as at the start. **Water is still the only thing being split.**

Contrast with what they have already seen:

Session	The dissolved salt	Its fate
2	CuSO_4	Cu^{2+} was consumed — it became the plating
3	AgNO_3	Ag^+ was consumed and slowly ran out
4	Na_2SO_4	Not consumed. It only carries charge

Ask: "If the sulfate is not used up, could we run this cell forever?" — until the water runs out, essentially yes. That is what an industrial electrolyzer does.

Why not table salt: chloride is easier to oxidize than water, so at the anode you would get **Cl_2 gas** instead of oxygen. This is not a rule for the sake of rules — it is a competing half-reaction, and it wins.

3.4 The two voltages

- **1.23 V** — the thermodynamic minimum. Below it, nothing, ever.
- **What you actually need** — usually **1.6–2.0 V** before you see anything, and more to go fast.

The gap is **overpotential**: an electrode surface can be thermodynamically willing and kinetically slow. Making an O_2 molecule requires assembling two oxygen atoms and breaking four O–H bonds on a surface that is not especially good at it, and that clumsiness costs voltage.

"This is why the platinum in a fuel cell is worth its price — not because it changes the chemistry, but because it makes the same chemistry faster at lower voltage."

3.5 Rate from current — Faraday, third time

Same law as Session 3, with the electron count changed:

Cathode (-): $2\text{H}_2\text{O} + 2\text{e}^- \rightarrow \text{H}_2 + 2\text{OH}^-$ 2 electrons per H_2
 Anode (+): $2\text{H}_2\text{O} \rightarrow \text{O}_2 + 4\text{H}^+ + 4\text{e}^-$ 4 electrons per O_2

$$\dot{n}(\text{H}_2) = I / (2F) \qquad \dot{n}(\text{O}_2) = I / (4F)$$

Say this clearly, it is the intellectual core of the day: the same current passes through both electrodes — it has nowhere else to go. So the number of electrons arriving at one is the number leaving the other. If hydrogen costs 2 electrons and oxygen costs 4, then **twice as much hydrogen must appear**. The 2:1 ratio is not a coincidence of this experiment; it falls out of counting.

Then convert to something they can see:

$$\dot{V}(\text{H}_2) \text{ in mL/min} \approx 7.6 \times I \qquad (I \text{ in amperes, room conditions})$$

Board this: *one amp $\approx$ 7.5 mL of hydrogen per minute.*

3.6 The flowchart (draw it)

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I (A)
→  $\dot{n}(\text{H}_2) = I/(2F)$  mol/s
→  $\times 24,450 \text{ mL/mol}$  → mL/s
→  $\times 60$  → mL/min
→  $\times t$  → PREDICTED VOLUME
      ↓
    compare with the U-tube
      ↓
measured  $\div$  predicted = current efficiency
  
```

That last box is what Session 3 could not do. Silver film was too thin to weigh; gas is easy to measure.

3.7 How the U-tube works (show the apparatus while you say this)

Point at the physical parts:

Part	Role
Stopper + graphite rod	Seals the top of each limb; gas collects underneath
The dome under the stopper	Where the gas accumulates
The side arm	The vent . Displaced liquid rises into it
Both limbs, same bore	So centimetres of column can be compared directly

The one instruction that decides success: *fill until the liquid stands part-way up both side arms*. Fill below them and the gas escapes out of the side arm, and the apparatus looks like it is working while collecting nothing.

3.8 Set expectations before they measure

Tell them **in advance** that the ratio will probably come out a little **above 2**, and why:

- Oxygen is about twice as soluble in water as hydrogen, so some of it quietly dissolves

- The graphite anode is slowly eaten, and some oxygen leaves as CO₂, which is very soluble indeed
- Oxygen evolution is slower to get going

Announcing this beforehand converts "our experiment was wrong" into "we predicted the direction of our own error." **A ratio below 2 is the one that means something is broken** — almost always a leak on the hydrogen side.

3.9 Safety (brief and firm)

- No table salt in any bath in this room today. Chloride makes chlorine
- Ammeter in series, never across the supply
- The U-tube is **never** sealed — side arms stay open
- Flame is instructor-only, and the balloon is untied only when it is well away from every cell that is still making hydrogen

4. Quiz questions (before the lab)

Cold call, pair-share, or a 3-minute written check. Answers in *italics*.

A. Review from Days 1–3

Q1. In Session 3, what did $Q = I \times t$ give us, and what did we divide it by to get moles of electrons? *Q was total charge in coulombs; divide by $F = 96,485 \text{ C/mol}$.*

Q2. Silver needed one electron per atom, copper needed two. Same charge through both cells — which gives more moles of metal? *Silver. Each atom is cheaper in electrons.*

Q3. Why was the graphite anode in Session 3 called "inert"? *It did not dissolve, so it did not resupply Ag^+ to the bath.*

B. Today's concepts

Q4. Pure water is 55.5 M in H₂O but only 10^{-7} M in H⁺. Roughly what fraction of molecules is ionized? *About one in 550 million — which is why it barely conducts.*

Q5. We add Na₂SO₄. Which of Na⁺, SO₄²⁻, or H₂O gets split? *Water. The sulfate is a spectator that only carries charge — it is there at the end, unchanged.*

Q6. So why bother adding it? *Without mobile ions, charge cannot cross the solution and no current flows, so no reaction happens at either electrode.*

Q7. Why is table salt forbidden here? *Chloride is oxidized more easily than water, so the anode would make Cl₂ gas instead of O₂.*

Q8. Write both half-reactions and say how many electrons each gas costs. $2\text{H}_2\text{O} + 2\text{e}^- \rightarrow \text{H}_2 + 2\text{OH}^-$ (2 e^- per H₂); $2\text{H}_2\text{O} \rightarrow \text{O}_2 + 4\text{H}^+ + 4\text{e}^-$ (4 e^- per O₂).

Q9. The same current flows through both electrodes. Explain in one sentence why that forces a 2:1 volume ratio. *Equal electrons at both electrodes, but hydrogen costs half as many per molecule, so twice as many H₂ molecules form — and equal numbers of gas molecules occupy equal volumes.*

Q10. A cell runs at 0.40 A. How many mL of H₂ per minute, roughly? $7.6 \times 0.40 \approx 3.0 \text{ mL/min}$.

Q11. Water needs 1.23 V. Your supply is set to 4 V. Is the extra 2.8 V wasted? *Largely yes — it pays for overpotential and solution resistance, and ends up as heat. It buys speed, not extra product per coulomb.*

C. Stretch

Q12. Your U-tube gives $V(\text{H}_2)/V(\text{O}_2) = 2.4$. Is the experiment wrong? *No — and the direction is predictable. O_2 is more soluble, and graphite anodes lose some oxygen as CO_2 . A ratio below 2 would be the worrying one, because it usually means the hydrogen side is leaking.*

Q13. You collect 82% of your predicted hydrogen. Where did the other 18% go? *Dissolved in solution, leaked past a stopper, or lost to side reactions on the graphite. This is exactly the current efficiency we had to assume was 100% in Session 3.*

5. Interactive activity + prediction sheet (required before Part C)

Activity name: "Predict the gas"

Format: pairs or groups of 3–4. **10–15 minutes.** **Goal:** every group writes down a **number in millilitres** before they collect anything — the direct sequel to Session 3's "Predict before you plate," except this time the prediction is checkable.

Part 1 — Station card

Parameter	Value
Planned current I	0.30 A (300 mA)
Planned collection time t	600 s (10 min)
Gas of interest	H_2 at the cathode (–)
U-tube internal diameter	2.0 cm

Part 2 — Live prompts

- Unit police (1 min).** Hold up a meter reading "300" on the mA range. "What number goes into the formula?" Chorus: **0.300 A**. Every year, someone divides by 2F using 300.
- The two-for-one (2 min).** "Same current at both electrodes. Hydrogen costs 2 electrons, oxygen costs 4. Which side fills faster, and by exactly how much?" Push until someone says it without hedging: **hydrogen, exactly twice**.
- Human flowchart (3 min).** Four volunteers hold signs: $I \rightarrow \dot{n} = I/2F \rightarrow \text{mL/min} \rightarrow \text{mL in 10 min}$. The class shouts what connects each pair.
- Order-of-magnitude bet (1 min).** Before any arithmetic: "Will 10 minutes at 300 mA give us a teaspoon, a soda can, or a bathtub?" Let them commit out loud. Then calculate. A teaspoon is 5 mL and the answer is about 23 mL — most classes badly overestimate, which is exactly the useful surprise.

Part 3 — Prediction sheet

Groups complete this before touching the U-tube. A TA initials one line per group.

Constants: $F = 96,485 \text{ C/mol e}^-$; molar volume $\approx 24,450 \text{ mL/mol}$ at 25°C

Step	Formula	Work	Answer + unit
1. Current in amperes	$\text{mA} \div 1000$		_____ A

Step	Formula	Work	Answer + unit
2. Mole rate of H ₂	$I / (2F)$		_____ mol/s
3. Mole rate of O ₂	$I / (4F)$		_____ mol/s
4. Check	rate H ₂ ÷ rate O ₂		should be 2
5. H ₂ volume rate	step 2 × 24,450 × 60		_____ mL/min
6. H ₂ in 10 min	step 5 × 10		_____ mL
7. O ₂ in 10 min	step 6 ÷ 2		_____ mL
8. Column height of H ₂	step 6 ÷ 3.14 mL/cm		_____ cm

Target answers for the card above (for you):

Step	Result
I	0.300 A
$\dot{n}(\text{H}_2)$	$0.300 / 192,970 \approx 1.55 \times 10^{-6} \text{ mol/s}$
$\dot{n}(\text{O}_2)$	$\approx 7.77 \times 10^{-7} \text{ mol/s}$
Ratio check	2.00
H ₂ rate	$\approx 2.28 \text{ mL/min}$
H ₂ in 10 min	$\approx 22.8 \text{ mL}$
O ₂ in 10 min	$\approx 11.4 \text{ mL}$
H ₂ column in a 2.0 cm tube	$\approx 7.3 \text{ cm}$

Sense-check to say: "About 23 mL — five teaspoons — after ten minutes. Splitting water is not hard, but it is slow. Remember that number when you read about hydrogen cars: an industrial electrolyzer is not doing anything cleverer than this, it is just doing it thousands of amps at a time."

Part 4 — Prediction statement (1 min)

Each group writes one sentence:

"Running at about __ A for _ minutes, we expect mL of hydrogen and ____ mL of oxygen, a ratio of 2:1."

After the run they recalculate with their **actual** current and time, and divide measured by predicted to get their current efficiency.

Gate before the U-tube

- [] Prediction sheet complete, ratio check equals 2
- [] TA has initialled the predicted volume
- [] Group can point to: cathode (–) = H₂ limb, anode (+) = O₂ limb, ammeter in series
- [] Group can state the fill rule: **liquid part-way up both side arms**

- ☐ Goggles on
-

6. During and after the experiment — what to emphasize

1. Make them **name the fast limb before reading the wires**. If the fast limb is not on (–), the leads are swapped and everything downstream is backwards.
 2. The 2–3 minute pre-saturation run before $t = 0$ is not optional. Groups that skip it get the worst ratios and blame the apparatus.
 3. When a group reports 2.3, resist calling it error. Ask: *"Which of our three reasons would push it that way?"*
 4. When a group reports 1.4, treat it as a fault to find: leaking stopper, or gas venting out of the side arm.
 5. Debrief questions:
 6. Did the sulfate get used up? How do you know?
 7. What was your current efficiency, and where did the missing gas go?
 8. Session 3 assumed 100% efficiency. Was that assumption safe?
 9. Where did the energy in the balloon pop come from?
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7. Bridge to Session 5

"Today electricity forced water apart — a reaction that would never happen on its own. Tomorrow we look at the opposite: a reaction that happens whether we want it or not. Iron rusting. And then, with the cells we are starting in the next ten minutes, we will have spent a night forcing that one backwards."

Start the derusting cells per experiment.md Part E, and make the whole class say the polarity out loud: **rusty iron on negative**.

8. Vocabulary checklist

- ☐ Electrolysis
 - ☐ Supporting electrolyte (vs a reactant salt)
 - ☐ Spectator ion
 - ☐ Overpotential
 - ☐ Current efficiency
 - ☐ Molar volume of a gas
 - ☐ Stoichiometry — as a volume ratio and as an electron count
 - ☐ Energy carrier vs energy source
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9. Materials you need in hand while teaching

- Beaker of **distilled** water with two graphite rods, pre-wired, for the opening hook
- The three ladder beakers, labeled
- A meter showing mA, for the unit-police moment
- Board space for the flowchart

- Printed prediction sheets (§5 Part 3)
- The U-tube, filled correctly, to point at while explaining §3.7
- Splints, and the balloon generator that has been running since before class

Session 4 — Instructor Notes: Water Electrolysis Deep Dive

Audience: instructors and TAs only. Pair with [lecture.md](#) and [experiment.md](#).

Why this session exists

For three days the students changed a **metal**. Today they change the **solvent**, and for the first time the product is something they can actually measure. That is what makes this the quantitative high point of the week: Session 3 asked them to trust Faraday's law, and Session 4 lets them audit it.

The session is built around one arc: **predict a volume from a current, then go and measure that volume**. Everything else — the conductivity ladder, the splint tests, the balloon — is scaffolding around that arc. If you are short on time, protect Parts B and C and cut elsewhere.

Teaching sequence (keep this order)

1. **Distilled water beaker** — expect disappointment. That disappointment is the lesson.
 2. **Ladder up through tap to electrolyte** — one voltage, three liquids, three currents.
 3. **Measure I and predict** a gas volume in millilitres, on paper, before collecting.
 4. **U-tube** — the ratio confirms the electron counting; the absolute volume audits the efficiency.
 5. **Splint tests** — cheap, reliable identification of both gases.
 6. **Balloon** — the memorable one, already running since before class.
 7. **Derusting cells** — handoff to Session 5.
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The three things that most often go wrong

Ranked by how much damage they do, and all three are avoidable:

1. The balloon generator started too late

This is the most common failure and it is pure arithmetic. Faraday sets the pace:

Current	H ₂ rate	Time to a 500 mL balloon
0.5 A	3.8 mL/min	132 min
1.0 A	7.6 mL/min	66 min
2.0 A	15.2 mL/min	33 min
3.0 A	22.8 mL/min	22 min

A "modestly inflated" party balloon is around 500 mL. There is no clever technique that beats this table — the electrons have to arrive one at a time. **Switch the generator on before the students walk in**, run the highest current your supply and electrodes tolerate, and pre-stretch the balloon so it does not fight the very low pressure the cell can deliver.

Pilot this and write the actual time in materials.md. It is the single most useful number to hand next year's version of yourself.

2. The U-tube filled below the side arms

Gas collects in the sealed dome under each stopper and pushes the liquid **down**; the displaced liquid rises into the open side arm, which is the vent. So:

- Fill until liquid **stands part-way up both side arms**, equal on both sides.
- If you fill to below the side-arm junction, the gas has a free path out of the side arm. The cell bubbles beautifully and collects nothing, which is a genuinely confusing failure to debug live.
- Stop the run before the falling level in the H_2 limb reaches its side-arm junction. The H_2 limb fills twice as fast, so it always gets there first, and once it vents the ratio slides toward 1.

3. Timing started before the solution is saturated

The first gas produced dissolves rather than collecting, and it does so unequally, because O_2 is roughly twice as soluble as H_2 . Run **2–3 minutes before marking $t = 0$** and most of that error disappears. Groups that skip it get the worst ratios and conclude the chemistry is unreliable.

Atomic and molecular foundations

Why pure water fails

Autoionization gives $[H^+] = [OH^-] = 10^{-7} M$ at 25 °C, against a water concentration of 55.5 M — about **one molecule in 550 million** is ionized. Conductivity of ultrapure water is around 0.055 $\mu S/cm$; typical tap water is 50–800 $\mu S/cm$; 0.1 M Na_2SO_4 is on the order of 10,000 $\mu S/cm$.

That spread is why the ladder is worth doing as three rungs rather than two. Tap water is the interesting middle case: it is emphatically not pure, and students who have been told "water doesn't conduct" need to meet the version of the statement that is actually true — *pure* water conducts poorly, and purity is a variable, not a property of the word "water."

Why Na_2SO_4 and not something else

Sodium is far too difficult to reduce in water (its reduction potential is well below that of water itself, so hydrogen evolves instead), and sulfate is already at maximum oxidation state, so neither ion has a competing half-reaction available. Both are pure charge carriers. This is the definition of a supporting electrolyte, and it is a concept students have not met in Sessions 2 and 3, where the dissolved salt was the *reactant*. Make the contrast explicit — the table in lecture.md §3.3 exists for this.

Electrolyte	Verdict	Reason
Sodium sulfate	Planned	Inert; no competing half-reaction
Baking soda	Backup	No chlorine risk, but CO_2 evolution muddies the anode gas measurement
Table salt	Never	Chloride oxidizes in preference to water $\rightarrow Cl_2$
Sulfuric acid	Not for this age group	Works well, but the hazard is not worth it

Half-reactions and where the 2:1 comes from

Cathode (-): $2\text{H}_2\text{O} + 2\text{e}^- \rightarrow \text{H}_2 + 2\text{OH}^-$ **Anode (+):** $2\text{H}_2\text{O} \rightarrow \text{O}_2 + 4\text{H}^+ + 4\text{e}^-$ (or $4\text{OH}^- \rightarrow \text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^-$ in base) **Overall:** $2\text{H}_2\text{O} \rightarrow 2\text{H}_2 + \text{O}_2$

Four electrons make **two H₂ and one O₂**. Series circuit $\Rightarrow$ identical charge through both electrodes $\Rightarrow$ the ratio is forced. Then Avogadro does the rest: equal numbers of gas molecules occupy equal volumes at the same temperature and pressure, so the mole ratio is the volume ratio.

Worth saying explicitly, because students often think the 2:1 comes from the formula H₂O. It does, but only via the electron count — and framing it that way is what makes Faraday's law feel like a tool rather than a formula.

Faraday coaching (Part B)

$$\dot{n}(\text{H}_2) = I / (2F) \quad \dot{n}(\text{O}_2) = I / (4F) \quad F \approx 9.65 \times 10^4 \text{ C/mol}$$
$$\dot{V}(\text{H}_2) \text{ in mL/min} \approx 7.6 \times I \quad (I \text{ in A, room conditions})$$

- **mA $\rightarrow$ A first, every time.** A thousandfold error is invisible on a calculator.
- The 2:1 is already present in the mole rates. The U-tube confirms chemistry students have *already derived*; it is not a new fact.
- Molar volume: 24.45 L/mol at 25 °C, or 22.4 L/mol at 0 °C. Pick one and stay with it all session. Mixing them introduces a 9% discrepancy that will surface exactly when you are computing efficiency and trying to explain a 15% shortfall.

The efficiency conversation

The measured-over-predicted ratio is the payoff of the whole week's arithmetic. Expect **70–95%**. Losses:

Loss	Typical size
Gas dissolved in solution	Largest single term, especially early
Leaks past stoppers	Variable; the fixable one
Side reactions on graphite (anode attack)	Small but real; grows with current
Gas escaping the side arm	Should be zero if the fill level is right

Above 100% is always a measurement artifact: an air bubble trapped under the stopper at $t = 0$, a mis-measured tube diameter (it enters squared, so a 10% diameter error is a 21% volume error), or a $t = 0$ mark made after the run had already started.

Voltage talking points

- 1.23 V thermodynamic floor.
- Onset of visible bubbling on graphite is typically **1.6–2.0 V** — the gap is overpotential, mostly on the oxygen side, which is kinetically the harder reaction.
- Several volts to run briskly; the surplus is dissipated as heat and buys rate, not yield per coulomb.
- If your supply is adjustable, hunting the bubbling threshold is a genuinely nice measurement and takes 90 seconds.

U-tube coaching (Part C)

Equal bore in both limbs means **column height ratio = volume ratio**, so the diameter is irrelevant to the ratio and only needed for the absolute Faraday check. Measure it once, put it on the board, done.

Accept 2.0–2.6. Tell students the expected direction *before* they measure — see lecture.md §3.8. Causes of a high ratio:

Cause	Note
O ₂ roughly 2× more soluble than H ₂	Dominant early; mostly cured by pre-saturating
Graphite anode attacked; some O as CO ₂	CO ₂ is highly soluble, so it vanishes from the gas phase
O ₂ evolution has the larger overpotential	Slower to start

A ratio **below 2** is a fault, not a subtlety. Check the H₂ stopper seal first, then whether the H₂ level reached the side arm.

Gas identification (Part D)

Splint tests — do these even if the balloon works

Gas	Test	Why it works
H ₂	Lit splint at the mouth → squeaky pop	H ₂ burns in the air around it; the pop is the flame front crossing the gas/air boundary
O ₂	Glowing splint → relights	Pure O ₂ raises the combustion rate enough to restart a smouldering ember

These take two minutes, cost nothing, work every time, and are a *better* piece of chemistry than the balloon because they identify **both** gases rather than one. At 0.3 A a 10 mL test tube of hydrogen fills in about 4 minutes — start them at the beginning of Part C.

Keep collection tubes **mouth-down** until the moment of testing. Hydrogen is 14 times lighter than air and leaves an upright tube almost immediately.

The balloon — safety and physics

Beyond the timing table above:

- **Cathode gas only.** A balloon of pure H₂ deflagrates: it burns at the surface where it meets air, giving a bark. A stoichiometric H₂/O₂ mixture **detonates** — much louder, genuinely dangerous, and never a classroom demo.
- **Never a rigid container.** A balloon fails by tearing; glass fails by becoming shrapnel.
- Untie and ignite **well away from the generator**, which is still making hydrogen the whole time.
- Warn the room. The bang startles, and some students find it genuinely distressing.
- Have the extinguisher present and know where it is, not merely "in the building."

The teaching moment is immediately afterwards, while the room is still noisy: that bang was the electrical energy of the last hour coming back out in a fraction of a second. Hydrogen stored it. That is what "energy carrier" means, and it is the honest version of the hydrogen-economy story — round-trip efficiency is well under 100%, and the electricity has to come from somewhere first.

Part E — Rust removal handoff

Role	Connection
Rusty iron	Battery – (cathode)
Sacrificial steel or graphite	Battery + (anode)
Electrolyte	Washing soda (preferred) or baking soda — never NaCl

Power supply matters more than anything else here. A 9 V alkaline holds about 500 mAh and a derusting cell pulls a few hundred mA. It is flat in roughly two hours, and Session 5 then opens on a nail indistinguishable from the control — which is a bad start to the synthesis day. Use a mains supply or a 5 V USB brick. If neither exists, run the cells for 60–90 minutes during class and accept a smaller effect, rather than trusting a battery to last the night.

Also: these cells evolve hydrogen continuously. Jars stay **open**, the room stays **ventilated**, and nothing sealed is ever used. Confirm the venue's overnight-power policy in advance rather than at 4 pm on the day.

Photograph the "before". Set up an unpowered control in the same solution. Without both, tomorrow's reveal is an anecdote.

Misconceptions

Misconception	Correction
"The bubbles are steam."	H ₂ and O ₂ . The water is being chemically split, not boiled — the beaker is barely warm.
"Water doesn't conduct electricity."	<i>Pure</i> water conducts poorly. Tap water conducts measurably. That is the whole point of the ladder.
"The salt gets used up making the gas."	Na ⁺ and SO ₄ ²⁻ are spectators. Water is the only reactant.
"Salt would work better."	Salt makes chlorine. A competing half-reaction, not a rule for its own sake.
"We need a big voltage because water is strong."	Minimum is 1.23 V. The excess is kinetics and resistance, and it becomes heat.
"Current and stoichiometry are separate topics."	Faraday's law is stoichiometry, written in electrons.
"The 2:1 comes from the formula H ₂ O."	It comes from the electron counts, 2 and 4 — which of course trace back to the formula, but the electrons are the mechanism.
"Hydrogen is a fuel like petrol."	It is a carrier. The energy came from the socket, and you never get all of it back.
"Our ratio was 2.3, so the experiment failed."	2.0–2.6 is the correct answer for this apparatus, and its direction is predictable.

Board plan

1. Beaker, two rods, distilled water. Write "1 in 550 million."

2. Add ions: draw Na^+ and SO_4^{2-} shuttling. Write "spectators."
 3. Half-reactions, with 2e^- and 4e^- circled.
 4. $\dot{n} = I/(nF) \rightarrow 7.6 \times I \text{ mL/min} \rightarrow$ predicted volume in a box.
 5. $2\text{H}_2\text{O} \rightarrow 2\text{H}_2 + \text{O}_2$, circle the 2:1.
 6. Reserve a corner for measured $\div$ predicted = efficiency. Fill it in live from the first group that finishes.
 7. Safety: no NaCl; side arms open; flame is mine.
 8. Before dismissal: **rusty iron on negative.**
-

Optional enrichment

- Conductivity measured properly, in $\mu\text{S/cm}$, if you have a meter — turns the ladder from qualitative to quantitative.
- Collected gas is saturated with water vapour; correcting for vapour pressure ($\sim 3.2 \text{ kPa}$ at 25°C , about 3% of atmospheric) is a nice touch for the strongest students and explains part of a systematic overestimate.
- Overpotential on graphite vs platinum, and why fuel cells pay for platinum.
- Alkaline vs PEM electrolyzers at industrial scale; the thousands-of-amps comparison lands well right after they compute 23 mL in ten minutes.
- **Universal indicator** in the electrolysis cell, if you can get it for next year: the cathode region turns blue-purple from OH^- and the anode region red-yellow from H^+ , making the half-reactions directly visible. Cheap, fast, and the most vivid single upgrade available to this session.

Session 4 — Preparation

The three things that decide whether this session works

1. **The balloon generator starts before the students arrive.** At a realistic classroom current a party balloon takes 30–60 minutes to fill. There is no way to hurry it.
2. **The U-tube is filled until liquid stands part-way up both side arms.** Filled any lower, the gas escapes out of the side arm and Part C silently collects nothing.
3. **The overnight derusting cells run on mains power, not a 9 V battery.** A 9 V alkaline is flat in about two hours, and Session 5 then opens on a nail that looks exactly like the control.

Everything else on this page is ordinary preparation.

Before class (instructor)

Pilot the measurements

- ☐ Run the **conductivity ladder** at a fixed voltage and record all three currents
- ☐ Work the **Faraday prediction** with your own pilot current — students will copy your method line for line
- ☐ Pilot the **U-tube**: confirm the fill level, and that a clear ratio appears inside 20 minutes
- ☐ **Measure the U-tube's internal diameter** and write it somewhere permanent
- ☐ Time the **balloon generator** at the current you intend to use, and record the minutes-to-usable in materials.md
- ☐ Rehearse both **splint tests** — the glowing-splint relight takes a little practice to get a splint to the right ember state

Solutions and hardware

- ☐ Mix ~**0.1 M Na₂SO₄** stock; check whether your salt is anhydrous (3.6 g/250 mL) or **decahydrate** (8.1 g/250 mL)
- ☐ Label every container "**0.1 M Na₂SO₄ — NO SALT**"
- ☐ Pre-weigh Na₂SO₄ portions for the Part A rung-3 addition
- ☐ One jug of **distilled water** for the ladder's first rung
- ☐ Check graphite rods, stoppers (**and spares**), spade connectors, multimeters, fresh meter fuses
- ☐ Confirm every multimeter's mA range actually works — a blown fuse reads zero and looks exactly like "no reaction"

Part E staging

- ☐ Rusty nails — **two per group**, matched in rustiness, so the control is a fair comparison
 - ☐ Washing soda, sacrificial anodes, open jars
 - ☐ **Mains supply or USB bricks** confirmed and counted
 - ☐ Overnight power policy confirmed **with the venue**, in writing if possible
 - ☐ Ventilation confirmed for the room the cells will sit in overnight
-

Day-of setup

- [] **T minus 5 minutes: switch on the balloon generator**
 - [] Ladder stations: three labeled beakers, graphite, leads, series ammeter
 - [] U-tubes filled to the correct level, side arms open, levels equal on both sides
 - [] U-tube internal diameter written on the board
 - [] Large sign: **"NO SALT IN ELECTROLYSIS — SODIUM SULFATE ONLY"**
 - [] Balloon area cleared and marked; extinguisher present; long lighter staged
 - [] Splints and collection tubes at each bench
 - [] Part E station ready to go at the 82-minute mark, not assembled from scratch then
 - [] Prediction sheets printed
-

Common failure modes

Issue	Prevention
"Part A is broken, nothing is happening"	Brief them first: a tiny current is the result
Meter reads zero after adding Na ₂ SO ₄	Wrong range, not in series, blown fuse, or bad clip — check the fuse first
Ladder currents not comparable	Electrode spacing or immersion depth changed between rungs. Fix the geometry, fix the voltage
U-tube collects nothing	Filled below the side arms. The most common single failure in this session
Ratio slides toward 1 late in the run	H ₂ level reached the side arm and is venting. Stop earlier
Poor ratios across the whole class	Nobody pre-saturated. Enforce the 2–3 min run before t = 0
Efficiency above 100%	Air trapped at t = 0, or a mis-measured diameter (it enters squared)
Balloon never fills	Started too late. Fall back to the splint tests and log the real fill time for next year
Students reach for the salt	Pre-brief; remove NaCl from the room entirely; only Na ₂ SO ₄ is on the benches
Derusting nails look untouched next morning	9 V battery died. Use mains power

Open questions / notes for next year

- Fixed voltage used for the ladder:
- Currents measured — distilled / tap / Na₂SO₄:
- Onset voltage for visible bubbling:
- U-tube internal diameter:
- Typical ratio and typical efficiency achieved by groups:
- Balloon generator current and actual fill time:
- Balloon demo permitted at this venue?
- Overnight power arrangement that was actually used: